Polite Bursting: Feedback Admission Control under Age-Limited Sensing on Shared Clusters

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***Abstract*—Research teams run batch GPU work on shared clusters they do not own. Fair-use policy caps how many pods a guest may hold and requires each pod to stay busy, so a guest has to restrain itself. How much capacity a guest can actually use is set by the other tenants. It is not announced, and a guest learns it only by probing. Every probe is already out of date by the time a controller acts on it. We show that acting on stale capacity readings has an exact ceiling. For a contended resource that is either free or taken, the extra goodput any feedback controller can win over the best fixed budget equals half the correlation between capacity now and capacity as last seen. A rule that admits work whenever the last reading showed the resource free attains that ceiling. The ceiling falls to zero once the sensing delay grows past the time over which neighbour load stays correlated, so past that point no feedback controller beats a fixed budget. We turn the result into two algorithms. The first estimates the correlation from its own probe stream and switches between feedback and a fixed budget, which holds it at or above the better of the two everywhere. The second admits work across several contended units by treating each unit as a separate binary decision, which extends the result beyond a single unit. Short controlled experiments support both. On a two-GPU node driven by a real competing tenant, feedback admission under structured neighbour load delivers 1.41 times the goodput of a fixed budget while over-admitting a quarter as often as a greedy policy, and it leaves the competing tenant more of its own throughput. Under memoryless load feedback only breaks even, which is what the ceiling predicts, and the switching algorithm detects that case and gives up the feedback loop rather than paying for it.**

I. Introduction

Much machine-learning work is bursty. Long stretches of local development are punctuated by short runs of batch jobs, sweeps, fine-tunes and evaluations, that briefly need far more GPUs than a team owns. Shared clusters such as the National Research Platform [1] make that capacity available on guest terms. Fair-use policy caps concurrent pods per user, requires sustained utilization of what is requested, forbids idle or sleep-only work, and reclaims resources on completion. The guest also sits outside the cluster network, often reachable only through one forwarded port or a mesh VPN, with no inbound access.

Two obvious strategies both fail here. Submitting everything at once takes more than a fair share, leaves pods pending or idle, and trips the thresholds that get a namespace banned. A fixed submission rate is safe but wastes the guest's own throughput whenever capacity is free. Neither one reacts to what the cluster is doing.

Being a good guest is a control problem, and how well it can be solved depends on how stale the guest's view of the cluster is. This paper makes that dependence exact and then builds admission algorithms on it.

Our first result is a ceiling on the value of feedback. When a controller acts on a capacity reading of a given age, the extra goodput it can win over the best fixed budget equals half the correlation between capacity now and capacity as last seen. The ceiling is tight, since a rule that admits work whenever the last reading showed capacity free reaches it. It also implies an impossibility. Once the sensing delay exceeds the time over which neighbour load stays correlated, the correlation is near zero and no feedback controller can beat a fixed budget.

Our second result turns the ceiling into algorithms. The correlation a controller needs is measurable from the probe stream the controller already collects, so a guest can estimate it online and act on the estimate. Algorithm 1 does this with one scalar of state, switching between a feedback loop and a fixed budget, and it never falls below the better of the two by more than a bounded amount. Algorithm 2 admits work across several contended units by running one binary decision per unit, which lifts the single-unit ceiling to any integer capacity.

Our third contribution is measurement rather than proof. The sensing delay is a physical quantity and we measure it at roughly 2.5 seconds on real hardware, dominated by CUDA start-up rather than by polling. We then drive a two-GPU node with a real competing tenant and show the predicted behaviour on both sides of the crossover, including a case where feedback stops paying for itself and the switching algorithm gives it up.

The control plane runs inside the cluster and reaches the home node over a single outbound port. The fabric that carries it is characterized in a companion paper [2] and is treated here as given, since what matters for the result is only that the guest's view of capacity arrives late.

II. Background and Motivation

Host policy. The reference host [1] publishes a fair-use policy that our design treats as ground truth. A user may hold at most $K{=}4$ concurrent pods. Per running pod, utilization relative to request must stay in published bands, at least 40% for GPU and 20 to 200% for CPU. Jobs must do real work, since sleep-only jobs are banned, and must release resources on completion. Idle storage volumes are reclaimed. Administrators monitor utilization and ban namespaces that hold under-utilized requests, so politeness is not optional.

Network reality. The guest reaches the cluster over one forwarded port or a mesh VPN, with no inbound access. A control plane must therefore live behind that boundary and

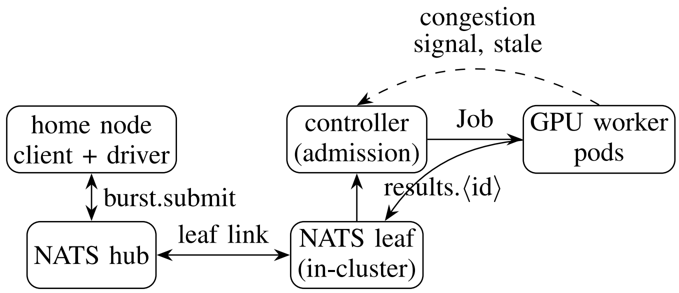


Fig. 1. PoliteBurst architecture. The home client submits over a federated NATS leaf link. The in-cluster controller admits jobs, pacing concurrency to the politeness budget, and launches GPU worker pods whose results route back through the leaf. The dashed arc is the stale feedback the controller acts on, and its delay bounds the achievable advantage by Thm. 1.

communicate over a fabric that traverses it, which motivates the leaf-node design of §VI.

Why existing tools do not settle it. Workflow and function-execution frameworks such as HTCondor [3], funcX and Globus Compute [4], Parsl [5], Ray [6] and Dask [7] scale workers to demand and carry no notion of a host politeness budget. Left to default scaling they over-provision under contention. We make self-restraint a controlled objective instead.

III. System Model

A guest bursts batch GPU pods onto a shared cluster under a fair-use policy that allows at most K concurrent pods and requires each admitted pod to sustain a utilization floor. Time is slotted into control epochs $t \in \mathbb{N}$. Let $a_t \in \{0, \dots, C\}$ be the capacity the cluster can productively give the guest at epoch t. It is set by the other tenants, so it is exogenous, stochastic and not directly observable. The guest chooses a window $w_t \in \{0, \dots, K\}$. Useful goodput is $G_t = \min(w_t, a_t)$, and the surplus $(w_t - a_t)^+$ is over-admitted, since those pods sit below the utilization floor and mark the guest impolite. Writing $\rho \triangleq \limsup_T \frac{1}{T} \sum_{t<T} \mathbf{1}[w_t > a_t]$ for the long-run over-admission fraction, the guest solves

$$\max_{\{w_t\}} \ \mathbb{E}[\min(w_t, a_t)] \quad \text{s.t.} \quad \underbrace{w_t \le K \ \forall t}_{\text{hard cap}}, \qquad \underbrace{\rho \le \epsilon}_{\text{policy}}. \tag{1}$$

The constraint is two-sided, since the guest must stay below the cap and above the floor, which already separates the problem from loss-only congestion control.

Age-limited sensing. The guest does not observe a_t. It probes cluster state and the probe is stale. We model this as an age of information D, meaning that at epoch t the controller's freshest reading of capacity has age D, so admissible policies are causal functions of $\{a_{s-D}\}_{s \le t}$. Here D is a physical and measured quantity. Sec. VIII puts it near 2.5 s, dominated by CUDA start-up rather than by the 27 ms poll. A controller can only act on a D-old picture of a cluster the neighbours are still changing.

A natural objection is that D is an engineering choice rather than a limit. Distributed schedulers routinely defeat stale probes by launching more work than they need and binding to whichever reservation is ready first, discarding the rest [8], or by holding a resident probe that reports at low latency. Under a one-sided constraint both work. Under the two-sided constraint of (1) neither is legal, and that is what makes D a floor rather than a parameter.

Proposition 1 (The sensing age is irreducible for a compliant guest)**.** *Let τ_{start} be the time from launching a unit of work until its productivity is observable. Any policy that satisfies both the cap $w_t \le K$ and the utilization floor has sensing age $D \ge \tau_{\text{start}}$ for the signal that reports whether an admission was productive.*

Proof. Reducing that age below τ_{start} requires one of three things. Observing the outcome of work not yet launched is impossible. Launching $m > 1$ speculative replicas and keeping the first to become productive consumes m slots against the cap and leaves $m-1$ admissions that perform no useful work, so it violates the floor whenever m exceeds the useful window. Holding a resident probe that is idle also violates the floor, and a resident probe doing real work consumes a cap slot while still reporting its own start-up at τ_{start}. Every route is therefore either infeasible or non-compliant. □

Proposition 1 is what separates this setting from load balancing on stale state, where the classical prescription is to sample few and avoid acting greedily on old readings [9], [10]. Those results hold under a one-sided constraint, where a guest that does nothing is compliant. Here a guest that holds a slot and does nothing is not, so the techniques that would shrink D are the ones the policy forbids. Everything that follows therefore treats D as given and asks what is achievable at that age.

The contended unit. Politeness is decided at the margin, so we fix attention on the single contended resource unit whose availability fluctuates with the neighbour. Let $x_t \in \{0, 1\}$ indicate that this unit is free ($x_t = 1$) or taken ($x_t = 0$), so admitting it when taken is the over-admission the policy punishes. We take x_t stationary with $\Pr[x_t = 1] = \frac{1}{2}$, a fully contended unit, and note that the asymmetric case only rescales the constants below. The guest's marginal decision $m_t \in \{0, 1\}$ is a possibly randomized causal function of the age-D observation $\hat{x}_t \triangleq x_{t-D}$. Marginal goodput and over-admission are

$$g \triangleq \Pr[m_t = 1, x_t = 1], \qquad \rho \triangleq \Pr[m_t = 1, x_t = 0]. \tag{2}$$

Everything below concerns the achievable (ρ, g) region under age D. The stationary ρ of (2) is the ergodic limit of the epoch fraction in (1), so we reuse the symbol, and write $\bar{\rho}$ for the surplus amount in Prop. 3.

Proposition 2 (Hard-cap safety)**.** *Any window update that clamps additive increase at K and never increases w_t on a congestion signal keeps $w_t \le K$ for every t deterministically, so the pod cap is never violated.*

Proof. Induction on t. Increase is clamped to K and decrease cannot exceed the current value. □

IV. How Much Stale Feedback Is Worth

We now characterize exactly how much closed-loop admission gains over a fixed budget when the only signal is a D-old observation. The answer is one interpretable quantity, the autocorrelation of capacity at the sensing lag.

Lemma 1 (Predictive disagreement)**.** *Let $q(D) \triangleq \Pr[x_t \neq x_{t-D}]$ be the lag-D disagreement probability and $r(D) \triangleq \mathbb{E}[(2x_t-1)(2x_{t-D}-1)]$ the lag-D autocorrelation of centered availability. Then $r(D) = 1-2q(D) \in [-1,1]$, and by stationarity and symmetry the joint law of $(\hat{x}_t, x_t)$ is $\Pr[\hat{x}_t{=}i, x_t{=}j] = \frac{1}{4}\big(1+(2i{-}1)(2j{-}1)\,r(D)\big)$. For a symmetric two-state Markov capacity with per-epoch flip probability $p \in (0,\frac{1}{2})$ we have $r(D) = (1-2p)^D = e^{-D/\tau_c}$ with coherence time $\tau_c \triangleq -1/\ln(1-2p) \approx 1/(2p)$.*

Proof. $r = \Pr[\text{agree}] - \Pr[\text{disagree}] = 1-2q$. The joint law follows from the two symmetric marginals and the single correlation. For the Markov chain, writing $\lambda_2 \triangleq 1-2p$ for the second eigenvalue, the D-step transition is $P^D = \frac{1}{2}\begin{pmatrix} 1+\lambda_2^D & 1-\lambda_2^D \\ 1-\lambda_2^D & 1+\lambda_2^D \end{pmatrix}$, so $\Pr[\text{disagree}] = \frac{1}{2}(1-\lambda_2^D)$ and $r = \lambda_2^D$. The two-state chain's autocorrelation and its total-variation distance to stationarity are standard mixing facts [11]. □

Theorem 1 (Achievable region and the feedback advantage)**.** *For the symmetric two-state Markov capacity of Lemma 1, under age-D sensing the achievable (ρ, g) region of* (2) *is the quadrilateral with vertices $\{(0,0), (\frac{q}{2}, \frac{1-q}{2}), (\frac{1}{2}, \frac{1}{2}), (\frac{1-q}{2}, \frac{q}{2})\}$ with $q = q(D)$, whose efficient upper boundary is the polyline $(0,0) \to (\frac{q}{2}, \frac{1-q}{2}) \to (\frac{1}{2}, \frac{1}{2})$. The best observation-oblivious policy is confined to the diagonal $g = \rho$. Consequently the goodput gain of feedback over a fixed budget, maximized over matched over-admission levels, is*

$$\boxed{\Delta(D) \;=\; \tfrac{1}{2}\,r(D) \;=\; \tfrac{1}{2}\big(1-2q(D)\big)} \tag{3}$$

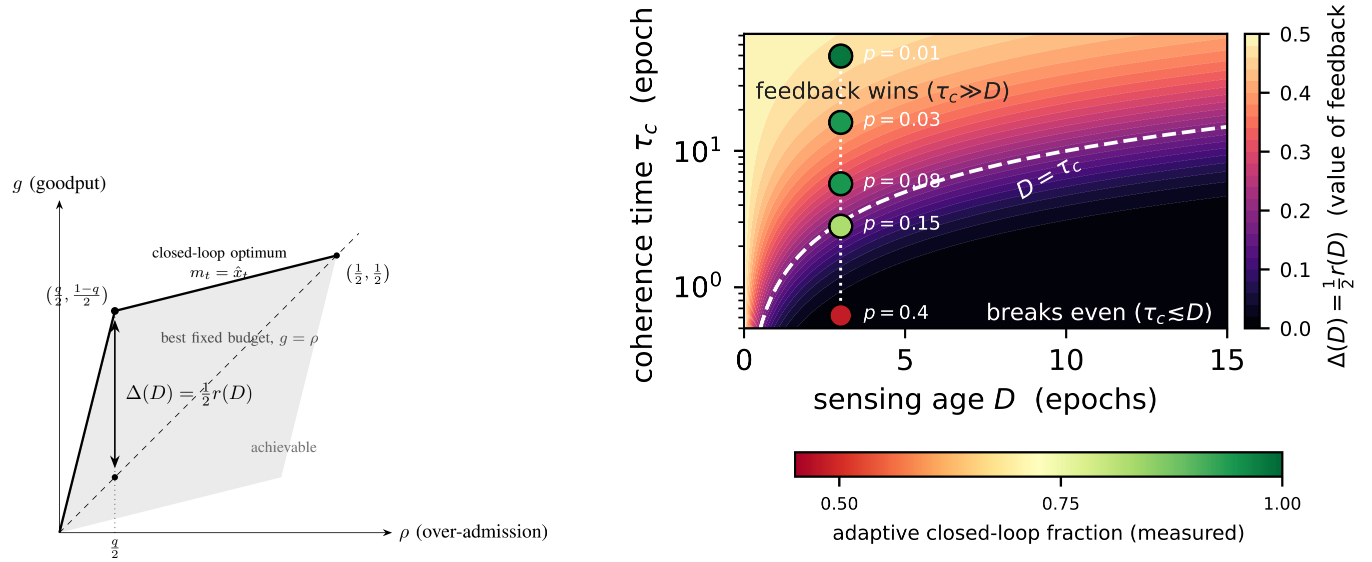
attained at the kink $\rho = q/2$ by the threshold policy $m_t = \hat{x}_t$, while at a lower level $\rho < q/2$ the gain is the smaller $\rho(1-2q)/q$. The peak value of closed-loop admission is therefore exactly half the autocorrelation of available capacity at the sensing lag (Fig. 2a). Optimality of memoryless threshold policies for symmetric two-state Markov observation is classical in opportunistic access [12], and what (3) *adds is the exact value of the resulting advantage over the best fixed budget.*

Proof. For the symmetric two-state Markov chain the freshest sample $\hat{x}_t = x_{t-D}$ is a sufficient statistic for x_t given the whole delayed history $\{x_{s-D}\}_{s \le t}$, since the Markov property makes older samples conditionally uninformative. An optimal causal age-D policy therefore depends on the history only through $\hat{x}_t$, so we may restrict without loss to policies parameterized by a pair (θ_1, θ_0) with $\theta_i = \Pr[m_t{=}1 \mid \hat{x}_t{=}i]$. Using Lemma 1, $g = \frac{1}{2}\big(\theta_1(1{-}q) + \theta_0 q\big)$ and $\rho = \frac{1}{2}\big(\theta_1 q + \theta_0(1{-}q)\big)$. This linear map sends the square $[0,1]^2$ to the quadrilateral with the stated vertices, whose upper boundary is the stated hull. Maximizing $g - \lambda\rho$ turns on θ_1 before θ_0 because $q < \frac{1}{2}$ makes the free-observation state strictly more goodput-efficient, so the maximum-goodput point for $\rho \le q/2$ is the kink $(\frac{q}{2}, \frac{1-q}{2})$, achieved by $m_t = \hat{x}_t$. A fixed policy ignores $\hat{x}_t$, so $\theta_1 = \theta_0 = \theta$ and $g = \rho = \theta/2$, the diagonal. The vertical distance at $\rho = q/2$ is $\frac{1-q}{2} - \frac{q}{2} = \frac{1-2q}{2} = \frac{1}{2}r(D)$. □

Corollary 1 (Impossibility as sensing ages)**.** $\Delta(D) \to 0$ *as $r(D) \to 0$. For the symmetric two-state Markov model $\Delta(D) = \frac{1}{2}e^{-D/\tau_c}$, so no causal controller with sensing age D outperforms a fixed budget once $D \gg \tau_c$. Closed-loop admission helps only to the extent that the guest senses the neighbour faster than the neighbour changes. In this model low autocorrelation and unpredictability coincide, but in general they need not, since a decorrelated process with $r(D){=}0$ can still be exactly predictable from its history, as a random-phase deterministic square wave is. For an arbitrary capacity process the vanishing of the advantage is therefore governed by the β-mixing coefficient rather than by $r(D)$. At $D = 0$ we have $\Delta = \frac{1}{2}$, and the advantage halves for every $\tau_c \ln 2$ epochs of added age (Fig. 2b). App. A makes the general statement precise and extends the impossibility to arbitrary non-binary and non-Markov capacity processes.*

Provenance, stated plainly. Equation (3) is elementary and we would rather derive it twice than appear to have hidden how short it is. For a symmetric binary hypothesis the Bayes error is $\frac{1}{2}(1 - \|\cdot\|_{\mathrm{TV}})$, so the largest advantage any test holds over blind guessing is exactly half the total variation between the two conditionals [13]. For a two-state chain that total variation is $|\lambda_2|^D$, which is $r(D)$. Compose the two and (3) follows in two lines. Each ingredient is old. Value of information is non-negative and convex in the prior [14] and is bounded by payoff oscillation times the movement of beliefs [15]. That movement at age D is the β-mixing coefficient [16], and the mixing facts for a two-state chain are standard [11]. Threshold optimality for positively correlated two-state availability is Ahmad et al. [12], and threshold optimality *under state-information lag* is Kim [17]. Measuring the value of information as degradation from the perfect-information gain is Brooks and Leondes [18]. Reducing a delay-D decision problem to an undelayed one by state augmentation is Katsikopoulos and Engelbrecht [19].

So the optimal policy is not ours and neither is the formula. Three things are. First, the statement is a *converse over every causal controller* at *matched over-admission*, not the value of one policy. The matched-constraint normalization is what makes "advantage over the best fixed budget" well posed at all, and it is the actual technical content of Thm. 1. Second, App. A lifts the converse off the binary Markov case to any stationary capacity process. Third, and most important for a systems venue, Prop. 1 shows D cannot be engineered away by a compliant guest, which is what makes an age-indexed limit the right object here rather than a modelling convenience. The exogenous structure also matters. Unlike the driven-plant setting of networked control, the admission action does not affect capacity, and that is what collapses the value of feedback onto a single predictability coefficient.



(a) the advantage is a vertical distance

(b) the impossibility as a map

Fig. 2. **(a)** Under age-D sensing the achievable (ρ, g) pairs form the shaded hull. Fixed budgets lie on the diagonal $g = \rho$, and the closed-loop optimum sits a vertical $\Delta(D) = \frac{1}{2}r(D)$ above it (Thm. 1). **(b)** The same value $\frac{1}{2}e^{-D/\tau_c}$ over the sensing-age and coherence-time plane with τ_c on a log scale. Points are the measured Markov contention sweep of Cor. 1, each shaded by the adaptive controller's closed-loop fraction, opening onto the fixed budget as D crosses τ_c.

Corollary 2 (The two contention profiles are two points on (3))**.** *The contention study of §VIII instantiates this two-state C=2 model. Structured square load has $\tau_c \gg D$, so $r(D)$ is near 1 and feedback is near-maximally useful. Memoryless Poisson load drives $\tau_c \lesssim D$, so $r(D)$ is near 0 and feedback collapses onto the fixed budget, which is the pre-registered null. The single curve* (3) *predicts both cases and the crossover between them.*

The binary reduction is not a special case but the atom of the general problem.

Proposition 3 (Layer-cake reduction for $C > 2$)**.** *Let $a_t \in \{0, \dots, C\}$ and score over-admission by the surplus amount $\bar{\rho} \triangleq \mathbb{E}[(w_t - a_t)^+]$. With $x_t^u \triangleq \mathbf{1}[a_t \geq u]$ and $\pi_u \triangleq \Pr[x_t^u = 1]$, both objectives decompose additively over units, since $\min(w_t, a_t) = \sum_{u \geq 1} \mathbf{1}[w_t \geq u]\, x_t^u$ and $(w_t - a_t)^+ = \sum_{u \geq 1} \mathbf{1}[w_t \geq u](1 - x_t^u)$. The age-$D$ window $w_t = a_{t-D}$ maximizes the advantage layer by layer, and its total goodput advantage over the best comonotone fixed budget at matched surplus obeys*

$$\Delta(D) \;\leq\; \sum_{u=1}^{K} \pi_u\, r_u(D), \qquad r_u(D) = \operatorname{corr}\big(x_t^u, x_{t-D}^u\big), \quad (4)$$

with equality if and only if $\pi_u\big(1 - r_u(D)\big)$ is non-increasing in u. For a single symmetric contended unit with $\pi_u = \frac{1}{2}$ this recovers $\Delta = \frac{1}{2}r(D)$, and equality also holds whenever $r_u(D)$ does not vary with u.

Proof. The two identities are the layer-cake expansions of $\min$ and $(\cdot)^+$. Setting $w_t = a_{t-D}$ makes $\mathbf{1}[w_t \geq u] = x_{t-D}^u$, which is the Thm. 1 threshold policy applied to each layer, and the layers' nesting $x_t^1 \geq x_t^2 \geq \cdots$ is inherited by the observations, so the per-unit thresholds compose into a valid scalar window. For an asymmetric layer with $\Pr[x^u{=}1] = \pi_u$, writing $P_{11}^u = \Pr[x_t^u{=}1, x_{t-D}^u{=}1]$, the same linear program as Thm. 1 gives per-unit advantage $\operatorname{Cov}(x_t^u, x_{t-D}^u)/(1 - \pi_u) = \pi_u r_u(D)$ over the fixed policy matching that layer's surplus, and additivity of goodput and surplus sums these to the right-hand side of (4).

The bound is an inequality rather than an identity because the summed comparator need not be realizable. Matching layer u's surplus requires the fixed policy to admit layer u with probability $\theta^u = \big(\pi_u - P_{11}^u\big)/(1 - \pi_u) = \pi_u\big(1 - r_u(D)\big)$, while any scalar window satisfies $\{w_t \geq u{+}1\} \subseteq \{w_t \geq u\}$ and therefore forces θ^u to be non-increasing in u. Since π_u decreases in u but $1 - r_u(D)$ may increase, the per-layer matched comparator is a feasible window exactly when $\pi_u(1 - r_u(D))$ is non-increasing, which gives the stated equality condition. When it fails, no single fixed budget matches every layer's surplus at once, the true comparator is strictly better than the layerwise one, and the sum overstates the advantage. □

At C=2 only the contended layer fluctuates, with π_1=1, r_1=0 and $\pi_2 = \frac{1}{2}$, so (4) collapses to $\frac{1}{2}r_2(D)$. The binary law is exactly the C=2 instance and the experiments measure it directly.

V. Two Algorithms

Theorem 1 leaves a shortfall that no fixed controller closes. Additive-increase and multiplicative-decrease tracks a slowly varying neighbour and approaches the kink when $\tau_c \gg D$, but over-admits when $\tau_c \lesssim D$, since it keeps probing a signal that has become noise. A fixed budget is safe everywhere

Algorithm 1 Adaptive admission for one contended unit

Require: cap K, floor $w_{\min}$, step α, backoff β, switch level θ, buffer length n, fixed budget $\bar{a}$

```
w ← w_min
for each control epoch t do
    append the newest probe reading to a buffer of the last n readings
    r̂ ← empirical correlation of readings D epochs apart
    if r̂ > θ then                          ▷ neighbour is trackable
        if the reading from D epochs ago showed over-admission then
            w ← max(w_min, ⌈βw⌉)
        else
            w ← min(w + α, K)
        end if
    else                ▷ not trackable, so stop paying for the loop
        w ← min(⌊ā⌋, K)
    end if
    admit w pods
end for
```

and forfeits $\Delta(D)$ when $\tau_c \gg D$. Which of the two is right depends on a quantity the controller can measure, because $r(D)$ is an autocorrelation of the probe stream the controller already collects.

Algorithm 1 estimates it online and switches. It holds one scalar of state beyond the AIMD window, costs constant time per epoch, and inherits Prop. 2, so it never exceeds the pod cap.

The tracking analysis of Algorithm 1 assumes **(A1)** stationary capacity $a_t \equiv a$ with $w_{\min} \le a \le K$, **(A2)** reliable detection of an overshoot within D epochs, and **(A3)** fixed $\alpha, \beta, w_{\min}, K$.

Proposition 4 (Sawtooth realization of the kink)**.** *Under stationary capacity a and (A1) to (A3), the trackable branch of Algorithm 1 forms a periodic sawtooth about a with long-run over-admission $\rho = D\alpha/\big((1-\beta)(a+D\alpha)\big) \approx D\alpha/((1-\beta)a)$. Choosing $\alpha \le \epsilon(1-\beta)a/D$ gives $\rho \le \epsilon$, and $\rho \to 0$ as $D \to 0$.*

Proof sketch. After a backoff the window restarts at $\beta(a + D\alpha)$ and rises by α per clean epoch, overshooting a for D epochs before detection triggers the next backoff at peak $a + D\alpha$. The fraction of violating epochs per cycle gives ρ. This is the dynamic approach to the static-capacity kink of Thm. 1, and D enters both here and in (3), which is the hinge between the systems measurement and the theory. □

Learning which branch to take has a sharp rate, and it is not the rate a generic chain concentration bound suggests. For the symmetric two-state chain the flip indicators $\mathbf{1}[a_t \neq a_{t-1}]$ are *exactly* independent Bernoulli(p) variables, since the flip probability is p from either state and after every history. Estimating p therefore carries no mixing penalty, and the relevant variance is $p(1-p)$ rather than the $1/\gamma$ proxy that a reversible-chain Hoeffding bound inserts. Since $\gamma = 2\min(p, 1-p)$, that true variance is about $\gamma/2$, so the spectral gap belongs in the numerator. We therefore use the Bernstein form of [20], which is stated in the asymptotic variance and is tight here.

Write p_θ for the solution of $(1-2p)^D = \theta$, let $s_\theta \triangleq 2D\,\theta^{(D-1)/D}$ be the sensitivity $|\mathrm{d}r/\mathrm{d}p|$ at that point, and let $\sigma_\theta^2 \triangleq p_\theta(1-p_\theta)$. Let $L > 0$ be the local slope in r of the difference between the two branches' values at θ, which is positive because Prop. 4 and Cor. 1 cross transversally there.

Theorem 2 (Matching learning rate)**.** *On the local family of symmetric two-state capacity processes at p_θ, the minimax* average per-epoch *shortfall against $\Delta(D)$ of any causal branch-selection rule over horizon N is*

$$\Theta\left(\frac{L\, s_\theta\, \sigma_\theta}{\sqrt{N}}\right) = \Theta\left(L\,\theta\sqrt{\frac{D\log(1/\theta)}{N}}\right), \tag{5}$$

and Algorithm 1 attains it. Over switch levels,

$$\sup_{\theta\in(0,1)} \Theta\left(\frac{L\, s_\theta\, \sigma_\theta}{\sqrt{N}}\right) = \Theta\left(L\sqrt{\frac{D}{N}}\right), \tag{6}$$

attained at $\theta = e^{-1/2}$. For a fixed process with $r(D) \neq \theta$ the shortfall is $O(1/N)$.

Proof sketch. For the upper bound, Bernstein for the independent flip indicators [20] gives $|\hat{p}_n - p| = O(\sigma_\theta/\sqrt{n})$ with high probability, and the delta method through the s_θ-Lipschitz map $p \mapsto (1-2p)^D$ gives $|\hat{r}_n - r(D)| = O(s_\theta\sigma_\theta/\sqrt{n})$. A wrong branch costs L times the correlation shortfall, and averaging over the misclassified prefix gives (5). For the lower bound, take two processes whose flip probabilities differ by δ. Their per-sample Kullback-Leibler divergence is $\delta^2/(2\sigma_\theta^2)$, so no rule separates them over N samples unless $\delta \gtrsim \sigma_\theta/\sqrt{N}$, which through the same map means correlations closer than $s_\theta\sigma_\theta/\sqrt{N}$ are indistinguishable. Le Cam's two-point argument [13] then forces average shortfall $\Omega(Ls_\theta\sigma_\theta/\sqrt{N})$, and both processes sit inside any class with a strictly smaller spectral gap, because their flip probabilities differ by $O(N^{-1/2})$. For (6), substituting $u = 1-2p$ gives $s_\theta\sigma_\theta = D\,u^{D-1}\sqrt{1-u^2}$, maximized at $u^2 = (D-1)/D$ where $\sqrt{1-u^2} = 1/\sqrt{D}$ and $u^{D-1} \to e^{-1/2}$, so the product is $e^{-1/2}\sqrt{D}$ and $\theta = u^D \to e^{-1/2}$. □

Two readings are worth stating. The spectral gap cancels, so no restriction to a class bounded away from zero is needed, and a rate written as $1/\sqrt{\gamma_0 N}$ would have the dependence inverted. And staleness costs only $\sqrt{D}$ in learning, which is far milder than its effect on the ceiling, where (3) decays exponentially in D. The hardest place to sit is $r(D) = e^{-1/2}$, neither clearly trackable nor clearly untrackable, which is exactly where a fixed controller has no good default and Algorithm 1 earns its state.

Proposition 3 lifts admission past a single unit. Algorithm 2 runs one binary decision per contended layer and adds them, which is a valid scalar window because the layers are nested, and its advantage is bounded by the per-unit sum (4) and attains it under the monotonicity condition there. It reduces

Algorithm 2 Layered admission for several contended units

Require: cap K, per-unit switch levels θ_u, per-unit fixed decisions $\bar{m}_u$, buffer length n

for each control epoch t **do**
    $w \leftarrow 0$
    for $u = 1$ to K **do** ▷ layers are nested, so $x_t^1 \geq x_t^2 \geq \cdots$
        $\hat{r}_u \leftarrow$ empirical lag-D correlation of layer u
        if $\hat{r}_u > \theta_u$ **then**
            $m_u \leftarrow \mathbf{1}$[layer u was free D epochs ago]
        else
            $m_u \leftarrow \bar{m}_u$
        end if
        $w \leftarrow w + m_u$
    end for
    admit $\min(w, K)$ pods
end for

to Algorithm 1 when only one layer fluctuates. The condition is worth stating operationally. Equality needs $\pi_u(1 - r_u(D))$ non-increasing in u, so it holds when every layer shares one correlation, as it does for a birth-death capacity process with a height-independent flip rate, and it can fail when the higher layers are markedly less predictable than the lower ones. Algorithm 2 is correct and safe either way, since Prop. 2 applies per layer. What the condition governs is only whether the achieved advantage reaches the summed ceiling.

Scope. Three assumptions earn the closed form. First, per layer the unit is binary and symmetric. For any symmetric stationary x_t the value $\frac{1}{2}r(D)$ is achievable by the memoryless threshold $m_t = \hat{x}_t$, which uses only the freshest sample, and it is the exact advantage when x_t is Markov, since then $\hat{x}_t$ is a sufficient statistic. For a general non-Markov symmetric process a history-using controller can do strictly better, as a random-phase square wave with $r(D)$=0 stays exactly predictable from its phase, so the equality is Markov-specific and the general converse is the mixing bound of App. A. Second, stationarity, since $r(D)$ is a lag-D autocorrelation of a stationary process. Under slow drift it is replaced by a local estimate, which is what Algorithm 1 already tracks. Third, fixed age D. If the freshest sample has a random age A the controller can read, the advantage is the age-averaged $\frac{1}{2}\,\mathbb{E}[r(A)]$, the same law under the age law, and the impossibility persists whenever $A \gg \tau_c$ almost surely.

VI. Realization: Sensing over a Federated Fabric

The controller runs inside the cluster, behind the network boundary, and reaches the home node over a NATS [21] leaf link that needs one outbound port (Fig. 1). It probes cluster occupancy and actuates admission across that link. Submissions flow on `burst.submit` and results return on `results.>` by interest propagation, with no inbound cluster access. What matters for the controller is that this view is late. The leaf round trip and the CUDA start-up of a freshly admitted pod together set the sensing age of §IV, which we measure in §VIII. Tensors crossing the link are compressed by random-rotation quantization [22], [23], which helps bandwidth on the hop but does not reduce the sensing age, since that age is dominated by CUDA start-up rather than transport. The fabric's own design, its leaf federation, durability and single-port traversal, is characterized in the companion paper [2].

VII. Implementation

PoliteBurst is a Go control plane with a decider, prober, submitter, NATS bridge and back-off modules, plus a Python client and an IPython cell magic [24]. The controller runs inside the cluster under a minimal namespace-scoped ServiceAccount holding Job create, get and delete rights and pod read, connects to the in-cluster NATS leaf, and renders each `JobDescriptor` into a Kubernetes Job. Node targeting through a `nodeSelector` on the descriptor pins guest pods to non-reserved GPU types. We verified the full path live on the reference host [1]. A submit from the home node propagated through hub, leaf and controller, launched a GPU worker that ran real CUDA work, and published its result back through the leaf to the home driver. The system, the controller and worker images, and the transport and right-sizing packages [22], [25] are available as open-source artifacts, with repositories anonymized for review.

VIII. Evaluation

We evaluate on a testbed whose sensing age is measured rather than assumed. A home node and a workstation-class node with GV100 GPUs run the controller, which probes real GPU occupancy and admits real CUDA workers while a competing tenant drives capacity. The fabric carrying this control plane is characterized in the companion paper [2]. Unless noted we report the mean and a t-based 95% confidence interval over independent repeats. Four questions organize the study, and each one defends a claim the theory makes. **(Q1)** What is the sensing age D that governs Prop. 4 and Prop. 1. **(Q2)** Does Algorithm 1 stay polite while beating a fixed budget under real time-varying contention, and where does it fail. **(Q3)** Does the value $\frac{1}{2}r(D)$ of Thm. 1 hold at the controller level, and does it also bind a learned controller.

A. Sensing age and the violation bound (Q1)

The sensing age has two measured parts. A read-only `nvidia-smi` probe costs 27 to 33 ms at the median. The end-to-end delay from launching a real GPU load until it is visible as utilization above the 40% floor is $\mathbf{2.46}$ s over 10 thermally guarded repeats on a GV100, with median 2.48 and range 2.29 to 2.61. CUDA start-up dominates that figure, not the poll. Which value applies depends on the question. Detecting whether a newly admitted pod is productive pays the full start-up cost, while probing pods that are already running costs only the poll. One epoch is a single completion-gated admission decision whose measured mean duration is about 1.1 s, so the 2.46 s start-up age is $D \approx 2$ to 3 epochs, which we sweep in Q3, and the 27 ms poll is below one epoch for running pods. Substituting into Prop. 4 makes D a first-order term in achievable politeness. A guest that admits aggressively with large α pays for it in over-admission scaled by this measured start-up delay.

B. Politeness under real contention (Q2)

We manufacture controlled contention on a two-GPU node with $C{=}2$. A competing tenant occupies $c(t) \in \{0, 1\}$ GPUs on a fixed profile, so the guest's available capacity $a(t) = C - c(t)$ is exogenous, time-varying and measured by the prober through a per-GPU process census subject to the sensing age of Q1. Two profiles bracket the space. *Square* load is a period-2τ wave modelling slowly varying or scheduled neighbours, and *poisson* load is memoryless on and off arrivals, the adversarial case. Alongside goodput and ρ we measure harm to the neighbour as the competitor's completed work relative to its undisturbed baseline. The study was pre-registered with randomized interleaved order and four seeds per cell.

Table I reports it. Under structured load Algorithm 1 is the polite and efficient knee. It reaches 1.41 times a fixed rate's goodput, 0.091 against 0.064 tasks/s with disjoint intervals, at $\rho = 0.15$, which is a quarter of naive's 0.64, and it spares the neighbour, retaining 76% of its throughput against naive's 62%. The target ϵ and switch level θ were fixed before the runs. Host policy penalizes persistently under-utilized requests rather than momentary dips, so ϵ between 0.1 and 0.2 is the tolerable share of under-utilized epochs, and the measured 0.15 sits inside it. Naive's higher goodput of 0.101 is bought entirely at the neighbour's expense, since it over-admits two-thirds of the time and takes 38% of the competitor's throughput.

Under memoryless load feedback breaks even and we report it. Goodput of 0.062 is statistically indistinguishable from the fixed rate's 0.064 with overlapping intervals, and over-admission is worse at $\rho = 0.30$, because capacity now varies on the timescale of the sensing delay. By the time the prober sees a competitor arrive the observation is stale. This is a pre-registered null and it is what Thm. 1 predicts, since the two profiles are two coherence times on the same curve (Cor. 2). Slow square load has $\tau_c \gg D$ and gives feedback room to help, while memoryless load drives $r(D)$ toward zero, where no causal controller beats a fixed budget. The break-even is therefore a consequence of the theorem rather than a tuning failure, and it is the case Algorithm 1 is built to detect. The measured sensing path also places the Prop. 4 sawtooth ceiling above the observed ρ in both profiles, which is conservative for a discrete completion-gated controller, and its prediction that politeness degrades as contention outpaces D is borne out. Naive is the most impolite in every case, at ρ between 0.64 and 0.85.

C. The value of feedback at the controller level (Q3)

The two contention profiles are two points, while Thm. 1 predicts the whole curve. To test it we simulate the window controllers on the two-state capacity process the theory analyzes, a Markov competitor with per-epoch flip probability

TABLE I
POLITENESS UNDER REAL CONTENTION WITH $C{=}2$, 4 SEEDS, MEAN WITH 95% INTERVAL. "KEPT" IS COMPETITOR THROUGHPUT RETAINED, WHERE 1 IS UNDISTURBED. UNDER SQUARE LOAD ALGORITHM 1 IS THE POLITE AND EFFICIENT KNEE, AND UNDER POISSON LOAD FEEDBACK BREAKS EVEN, A PRE-REGISTERED NULL.

contention	policy	goodput (tasks/s)	ρ	kept
square	naive	0.101 ± 0.002	0.64 ± 0.01	0.62
	aimd	0.091 ± 0.000	0.15 ± 0.01	0.76
	static	0.064 ± 0.000	0.11 ± 0.01	0.89
poisson	naive	0.096 ± 0.002	0.85 ± 0.06	0.62
	aimd	0.062 ± 0.006	0.30 ± 0.06	0.55
	static	0.064 ± 0.000	0.22 ± 0.04	0.79

p so the coherence time τ_c sweeps continuously, at the measured sensing age $D{=}3$ epochs with 1.2×10^5 epochs per point. Fig. 3 reports each controller's goodput advantage over the best fixed budget at matched over-admission against the predicted $\frac{1}{2}r(D)$.

Four readings confirm the theory and motivate Algorithm 1. First, the threshold policy sits on the predicted line across three decades of τ_c, with maximum deviation 0.013, so the closed form is the exact frontier rather than an asymptote. Second, fixed AIMD captures roughly half the available advantage when $\tau_c \gg D$ and goes negative, meaning worse than a fixed budget, once $\tau_c \lesssim D$, because it keeps paying over-admission for a signal that has become noise. That is the same failure the live poisson case shows. Third, Algorithm 1, estimating $r(D)$ online, tracks the upper envelope. It never drops below a fixed budget and recovers most of AIMD's advantage where feedback is worth having. Fourth, the value binds a learned controller. A policy-gradient agent in the DeepRM family [26], using a tabular softmax with a per-state baseline [27] and trained on the age-D observation history of four delayed readings, which is strictly more information than the memoryless threshold uses, converges to the threshold policy's advantage within 0.003 at every point and exceeds $\frac{1}{2}r(D)$ by at most 0.013, the threshold policy's own sampling deviation. Learning finds the frontier and does not pass it, which is the ceiling Thm. 1 sets for that family [26], [28]. Sweeping D collapses every curve onto $\frac{1}{2}e^{-D/\tau_c}$ against D/τ_c (Fig. 3, right), confirming the predicted scaling. The live square and poisson cases of Q3 are the two ends of that axis.

Live confirmation on GPU hardware. We ran Algorithm 1 on the same two-GPU node across the Markov τ_c sweep, with p from 0.01 to 0.4, $D{=}3$ and two seeds, estimating $r(D)$ online. Two claims transfer from the model. The online estimate lands on the predicted curve, with $\hat{r} = (1 - 2\hat{p})^D$ tracking $(1 - 2p)^D$ to within 0.07 (Fig. 4, left). And the switch modulates as predicted, its closed-loop fraction falling from 0.98 to 0.48 as $r(D)$ drops past θ. Hardware differs from simulation in one instructive way. Raw goodput rewards over-admission, so live AIMD out-performs a fixed budget everywhere, and at high flip rate it buys that goodput with impoliteness at $\rho = 0.44$ against 0.04 rather than by tracking.

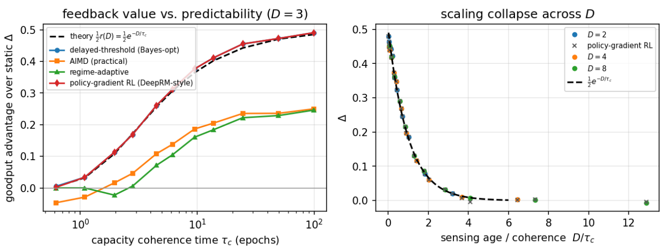


Fig. 3. Controller-level validation. *Left.* Goodput advantage over a fixed budget against capacity coherence time τ_c at D=3. The threshold policy lies on the predicted $\frac{1}{2}r(D)$, dashed, to within 0.013, and the DeepRM-style policy-gradient agent, diamonds, converges onto it without passing it. Fixed AIMD falls below a fixed budget once $\tau_c \lesssim D$, and Algorithm 1 tracks the envelope. *Right.* Sweeping D collapses the advantage onto $\frac{1}{2}e^{-D/\tau_c}$ against D/τ_c.

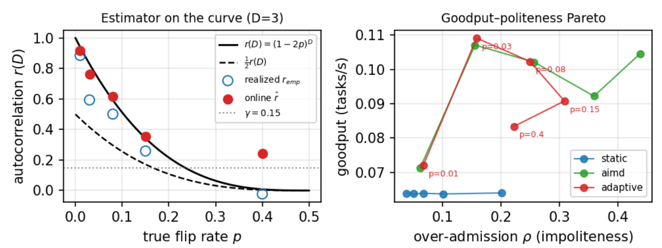


Fig. 4. Algorithm 1 live on two GV100 GPUs, Markov τ_c sweep, D=3, two seeds. *Left.* The online estimate $\hat{r}$ and the realized availability autocorrelation land on $(1-2p)^D$, and the switch fires as $r(D)$ crosses θ. *Right.* The goodput and over-admission frontier, where the adaptive controller tracks AIMD when feedback is useful and moves toward a fixed budget's politeness as p grows.

The predicted crossover therefore does not appear in raw goodput. It appears on the goodput and politeness frontier (Fig. 4, right) as the over-admission the switch sheds against always running AIMD, which grows from $+0.01$ to $+0.22$ as p rises from 0.08 to 0.4. Algorithm 1 matches AIMD where the neighbour is trackable and recovers a fixed budget's politeness where it is not.

D. Summary and threats to validity

The evaluation establishes three things. The sensing age is a measured first-order term at about 2.5 s, dominated by CUDA start-up, which is what Prop. 1 needs. Under structured contention Algorithm 1 is the polite and efficient knee and spares the neighbour, while under memoryless contention feedback breaks even, a pre-registered null we report rather than suppress. And the value $\frac{1}{2}r(D)$ holds at the controller level, with the threshold policy on it to within 0.013, a learned policy-gradient agent capped by it, and Algorithm 1 never below a fixed budget, which unifies the two live contention cases as endpoints of one curve.

Three threats remain. Contention is manufactured on a two-GPU node at C=2 with synthetic profiles rather than drawn from a production multi-tenant trace. The clean tracing of the advantage in Q4 is controller-level, because live static is a fixed trickle with no tunable frontier to measure an advantage against. And the live adaptive sweep uses two seeds per point, so it is a confirmatory anchor rather than primary evidence. The full system path is nonetheless verified end-to-end in §VII, and naive's impoliteness, at $\rho \geq 0.64$ with at least 38% neighbour harm, is robust across every case.

IX. Related Work

Bursting and function execution. HTCondor [3], funcX and Globus Compute [4], Parsl [5], Ray [6] and Dask [7] provide elastic task execution and provider-driven scaling, and treat the host as owned capacity to fill. None of them models a fair-use politeness budget or admission under a delayed view of capacity. They are complementary substrates, and a head-to-head comparison against them appears in the companion systems paper [2].

Congestion and scavenger control. Our loop is AIMD in the lineage of additive-increase and multiplicative-decrease congestion avoidance [29], and its objective of yielding to incumbents and using only slack aligns with low-priority transports such as LEDBAT [30] and TCP Nice [31]. Those protocols back off on one-sided loss or delay signals. Two things differ here. The objective is two-sided, since the guest must stay under a pod cap and above a utilization floor, which admits an over-admission measure rather than pure loss. And we bound what any such rule can achieve rather than proposing one more back-off schedule.

Learned resource control. Deep reinforcement learning schedulers including DeepRM [26] and Decima [28], with variance reduction for exogenous input processes [27], learn scheduling policies from cluster feedback. Our ceiling applies to them unchanged, since it constrains any causal policy rather than any particular family, and §VIII-C tests a policy-gradient agent of that family against it.

Age of information. The ceiling is a statement about freshness, since the value of feedback is set by the age of the capacity estimate relative to how fast the neighbour changes. That places polite bursting in the age-of-information line of work [32], on the control side rather than the sampling side. We bound the achievable goodput and politeness frontier as a function of sensing age and give algorithms that adapt to the measured value.

X. Conclusion

Restraint on a shared cluster is a control problem whose difficulty is fixed by how stale the guest's view of the cluster is. We gave that dependence an exact form. The extra goodput a feedback controller can win over the best fixed budget is half the correlation between capacity now and capacity as last seen, which is attained by admitting work whenever the last reading showed capacity free, and which vanishes once the sensing delay outgrows the neighbour's correlation time.

Two algorithms follow from that form. One measures the correlation online and switches between feedback and a fixed budget, staying at or above the better of the two. The other runs a binary decision per contended unit and so covers capacities beyond a single unit. Both carry the safety property that the pod cap is never exceeded, which holds deterministically rather than in expectation.

Short controlled experiments support the result on both sides of the crossover. Under structured neighbour load, feedback admission reaches 1.41 times a fixed budget's goodput at a quarter of a greedy policy's over-admission while leaving the competing tenant more throughput. Under memoryless load it breaks even, and the switching algorithm detects the case and stops paying for a loop that cannot help. A learned policy-gradient controller given a longer observation history converges to the ceiling and does not pass it.

Three things remain open. Our contention is manufactured on a two-GPU node rather than drawn from a production multi-tenant trace. The exact equality holds for a binary Markov unit, while the general statement is the weaker mixing bound of App. A. And a controller that estimates the correlation itself, rather than only which side of the switch it is on, should recover part of the remaining distance to the ceiling.

Appendix A
General-Capacity Impossibility via β-Mixing

Corollary 1 showed that feedback's value vanishes with the sensing age D for the two-state model. We now show the same for any stationary capacity process, with the decay rate governed by its β-mixing coefficient, also called absolute regularity. The exact binary law is one tight instance of that general statement.

Setup. Let $\{a_t\}$ be stationary on $\{0,\ldots,C\}$ with $\mathcal{F}_{\le s} = \sigma(a_u : u \le s)$ and $\mathcal{F}_{\ge s} = \sigma(a_u : u \ge s)$. Its β-mixing coefficient at lag D is

$$\beta(D) = \mathbb{E}\left[\sup_{B\in\mathcal{F}_{\ge t}} \left|\mathbb{P}(B \mid \mathcal{F}_{\le t-D}) - \mathbb{P}(B)\right|\right], \qquad (7)$$

and $\beta(D) \to 0$ holds if and only if the process is absolutely regular. Geometrically mixing processes satisfy $\beta(D) \le c_0\varrho^D$ for some $\varrho \in (0,1)$. Note that $\beta(D)$ always carries its lag argument and is unrelated to the AIMD back-off factor β. Fix any bounded per-epoch objective $R(w,a)$, for instance the politeness Lagrangian $R_\lambda(w,a) = \min(w,a) - \lambda(w-a)^+$, with oscillation $\Omega \triangleq \max_{w,a} R - \min_{w,a} R$. A causal age-$D$ controller sets w_t measurable with respect to $\mathcal{G} \triangleq \mathcal{F}_{\le t-D}$, while a fixed controller uses a $\mathcal{G}$-independent w. Write $V^\star(D)$ and V_{stat} for their optimal per-epoch values and $\Delta(D) = V^\star(D) - V_{\text{stat}}$.

Lemma 2 (Lipschitz value function)**.** *$\psi(\mu) \triangleq \max_w \sum_a R(w,a)\,\mu(a)$ is convex on the simplex and Ω-Lipschitz in total variation, so $|\psi(\mu)-\psi(\mu')| \le \Omega\,\|\mu-\mu'\|_{\text{TV}}$. Theorem 3 below is then the standard value-of-information bound [14], [15] specialized to the mixing coefficient.*

Proof. ψ is a pointwise maximum of linear functionals, hence convex. Let w_μ attain $\psi(\mu)$. Then $\psi(\mu) - \psi(\mu') \le \sum_a R(w_\mu,a)\,(\mu-\mu')(a) \le \operatorname{osc}_a R(w_\mu,\cdot)\,\|\mu-\mu'\|_{\text{TV}} \le \Omega\|\mu-\mu'\|_{\text{TV}}$, using $\sum_a f(a)\nu(a) \le \operatorname{osc}(f)\|\nu\|_{\text{TV}}$ for zero-sum ν. The reverse bound is symmetric. □

Theorem 3 (β-mixing impossibility)**.** *For any stationary capacity process and any bounded objective,*

$$0 \;\le\; \Delta(D) \;\le\; \Omega\,\beta(D). \qquad (8)$$

Hence no causal age-D controller outperforms a fixed budget as $D \to \infty$ whenever the capacity is absolutely regular. If $\beta(D) \le c_0\varrho^D$ the advantage decays geometrically, with $\Delta(D) \le \Omega c_0 \varrho^D$.

Proof. Let $\mu_{\mathcal{G}} = \mathbb{P}(a_t \in \cdot \mid \mathcal{G})$, so the tower property gives $\mathbb{E}[\mu_{\mathcal{G}}] = \bar\mu \triangleq \mathbb{P}(a_t \in \cdot)$. Optimizing the $\mathcal{G}$-measurable window pointwise, $V^\star(D) = \mathbb{E}[\psi(\mu_{\mathcal{G}})]$ and $V_{\text{stat}} = \psi(\bar\mu)$, so convexity of ψ and Jensen give $\Delta(D) = \mathbb{E}[\psi(\mu_{\mathcal{G}})] - \psi(\bar\mu) \ge 0$, meaning fresher information never hurts. For the upper bound, Lemma 2 gives $\Delta(D) \le \Omega\,\mathbb{E}\|\mu_{\mathcal{G}} - \bar\mu\|_{\text{TV}} = \Omega\,\mathbb{E}[\sup_B |\mathbb{P}(a_t{\in}B \mid \mathcal{G}) - \mathbb{P}(a_t{\in}B)|] \le \Omega\beta(D)$. The last step holds because $\sigma(a_t) \subseteq \mathcal{F}_{\ge t}$, so (7) takes the supremum over a larger field. □

Corollary 3 (Binary tightness)**.** *For the symmetric two-state model, $\beta(D) = \frac{1}{2}|1-2q(D)| = \frac{1}{2}r(D)$ by a standard mixing identity [16], [11], so Thm. 3 reproduces the exact advantage of Thm. 1 up to the objective scale Ω.*

Proof. The Markov property reduces $\mathcal{F}_{\le t-D}$ to a_{t-D}. The conditional-to-marginal total-variation distance is $|\frac{1}{2} - q(D)|$ from either state, so $\beta(D) = \frac{1}{2} - q(D) = \frac{1}{2}r(D)$. □

Appendix B
The Learning Rate of Algorithm 1

We give both directions of Thm. 2. Observations are delayed but clean, so $\hat a_i = a_{i-D}$, and Algorithm 1 forms $\hat p_n = \frac{1}{n}\sum_{i\le n} \mathbf{1}[\hat a_i \ne \hat a_{i-1}]$ and $\hat r_n = (1-2\hat p_n)^D$.

Since $\Pr[a_t \ne a_{t-1} \mid a_0,\ldots,a_{t-1}] = p$ for every history, $n\hat p_n$ is exactly Binomial(n,p), with variance $p(1-p)/n$ and per-sample Fisher information $1/(p(1-p))$. A generic reversible-chain Hoeffding bound would replace that variance by a range proxy of order $1/\gamma$, overstating the sample requirement by order $1/\gamma^2$ and inverting the sign of the γ dependence, so we use the Bernstein form of [20], which is stated in the asymptotic variance and is exact here.

Upper bound. Bernstein gives $|\hat p_n - p| = O(\sigma_\theta\sqrt{\log(1/\delta)/n})$ with probability $1-\delta$. The map $p \mapsto (1-2p)^D$ has derivative s_θ at p_θ, so $|\hat r_n - r(D)| = O(s_\theta\sigma_\theta\sqrt{\log(1/\delta)/n})$. A wrong branch costs at most L times the shortfall in r, and summing while n grows, then dividing by the horizon, gives $O(Ls_\theta\sigma_\theta/\sqrt{N})$.

Lower bound. Take two processes with flip probabilities $p_\theta \pm \delta/2$. Independence of the flip indicators makes the Kullback-Leibler divergence between n-sample laws $n\delta^2/(2\sigma_\theta^2) + O(n\delta^3)$, so no test separates them with constant probability unless $n \gtrsim \sigma_\theta^2/\delta^2$. Their correlations differ by $s_\theta\delta$ and the loss is locally *linear* in $r-\theta$ at the indifference level, so a rule that cannot distinguish them pays $\Omega(Ls_\theta\delta)$ on one. Le Cam's two-point argument [13] with $\delta = \sigma_\theta/\sqrt{N}$ matches the upper bound. Local linearity pins the exponent at $N^{-1/2}$, where a loss quadratic in $r-\theta$ would give N^{-1}. The two flip probabilities differ by $O(N^{-1/2})$, so their spectral gaps agree to that order and the pair certifies the rate without invoking a class boundary. That matters because θ is fixed by the objective, so on the hard manifold $r(D) = \theta$ the quantities D, $\gamma_\theta = 1 - \theta^{1/D}$ and s_θ are mutually determined. Treating them as free is what yields a spurious $1/\sqrt{\gamma_0 N}$ rate.

Worst switch level. With $u = 1 - 2p$ and $\theta = u^D$ we have $s_\theta \sigma_\theta = D\, u^{D-1}\sqrt{1-u^2}$. Setting the derivative of $(D-1)\log u + \frac{1}{2}\log(1-u^2)$ to zero gives $u^2 = (D-1)/D$, so $\sqrt{1-u^2} = 1/\sqrt{D}$ and $u^{D-1} \to e^{-1/2}$. The product is $e^{-1/2}\sqrt{D}$ at $\theta \to e^{-1/2}$, which is (6).

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